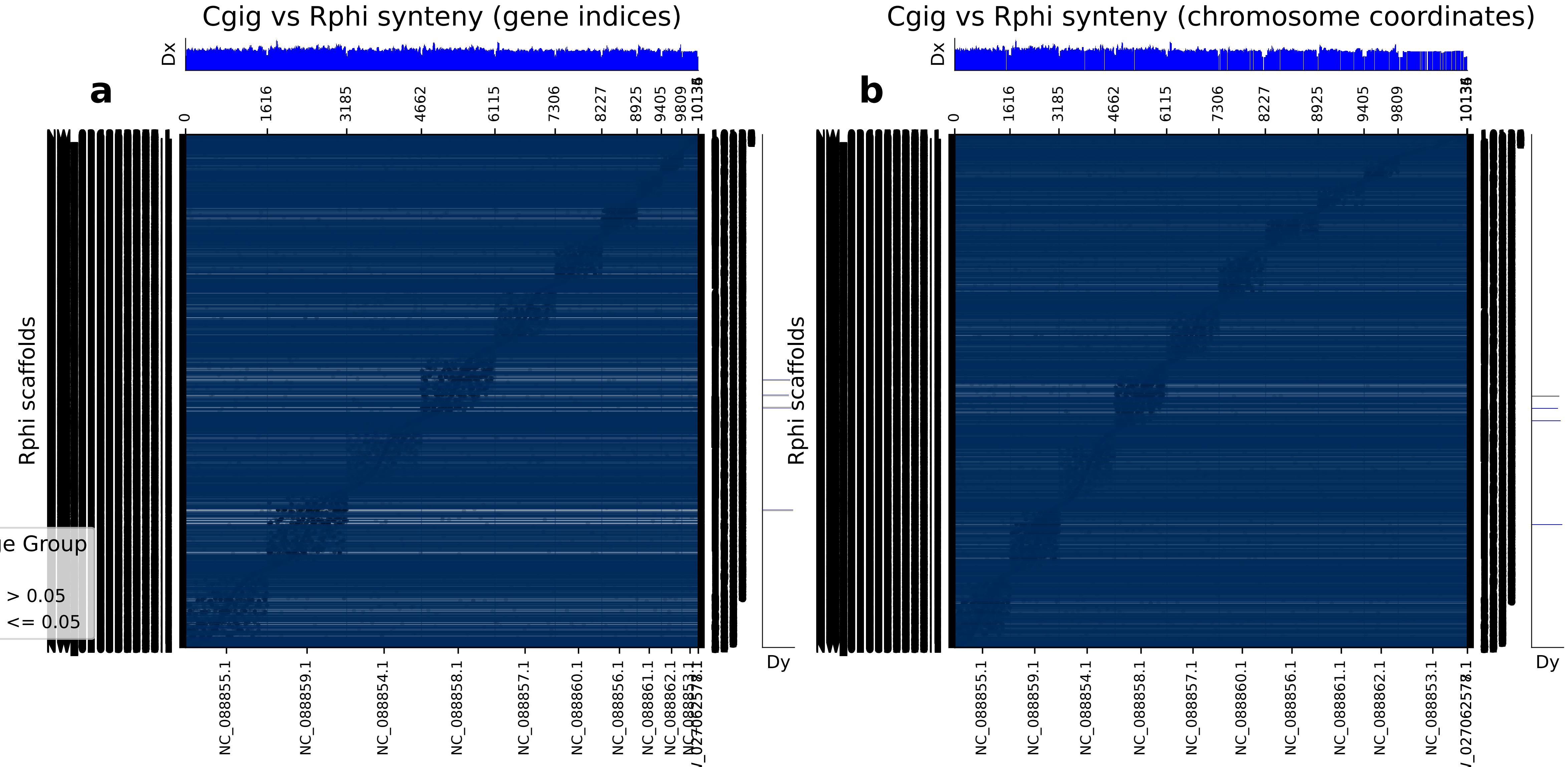
diamond: 10137 orthologs on 12 Cgig scaffolds and 4167 Rphi scaffolds.



c

Cgig vs Rphi groups

Cgig scaffolds

d

Cgig vs Rphi FET, whole chroms.

	Cgig scaf	Rphi scaf	p (FET)	Count
0	NC_088859.1	NW_026860903.1	4.30e-13	21
1	NC_088858.1	NW_026865571.1	1.09e-11	20
2	NC_088858.1	NW_026865484.1	6.64e-10	19
3	NC_088858.1	NW_026861563.1	7.31e-11	19
4	NC_088859.1	NW_026862663.1	1.66e-08	18
5	NC_088858.1	NW_026861019.1	2.57e-08	18
6	NC_088860.1	NW_026851633.1	1.49e-11	16
7	NC_088859.1	NW_026853802.1	5.08e-09	16
8	NC_088859.1	NW_026852008.1	4.54e-07	15
9	NC_088858.1	NW_026852258.1	1.06e-06	15
10	NC_088856.1	NW_026858503.1	1.61e-13	15
11	NC_088855.1	NW_026863896.1	2.63e-05	15
12	NC_088855.1	NW_026855197.1	1.06e-04	15
13	NC_088859.1	NW_026865967.1	2.77e-06	14
14	NC_088859.1	NW_026859933.1	9.22e-05	14
15	NC_088859.1	NW_026854960.1	2.16e-07	14
16	NC_088859.1	NW_026852705.1	2.77e-06	14
17	NC_088858.1	NW_026863541.1	7.32e-08	14
18	NC_088859.1	NW_026853170.1	1.40e-06	13
19	NC_088857.1	NW_026859561.1	3.84e-08	13
20	NC_088855.1	NW_026861864.1	2.48e-05	13
21	NC_088855.1	NW_026861084.1	1.59e-04	13
22	NC_088854.1	NW_026852000.1	6.37e-07	13
23	NC_088859.1	NW_026862783.1	1.02e-04	12
24	NC_088858.1	NW_026859790.1	3.62e-06	12
25	NC_088855.1	NW_026855294.1	3.68e-02	12
26	NC_088860.1	NW_026863645.1	1.82e-06	11
27	NC_088859.1	NW_026867092.1	5.93e-05	11
28	NC_088858.1	NW_026853905.1	2.54e-05	11
29	NC_088857.1	NW_026861840.1	7.33e-04	11
30	NC_088856.1	NW_026867313.1	7.66e-09	11
31	NC_088855.1	NW_026857770.1	1.88e-02	11
32	NC_088855.1	NW_026857221.1	8.48e-04	11
33	NC_088859.1	NW_026866451.1	3.64e-03	10
34	NC_088859.1	NW_026864352.1	1.88e-02	10
35	NC_088859.1	NW_026859998.1	3.85e-04	10
36	NC_088859.1	NW_026859757.1	3.85e-04	10
37	NC_088858.1	NW_026866837.1	1.71e-03	10
38	NC_088858.1	NW_026862689.1	1.71e-03	10
39	NC_088858.1	NW_026859620.1	1.71e-03	10
40	NC_088858.1	NW_026859418.1	1.78e-04	10
41	NC_088858.1	NW_026852468.1	1.71e-03	10
42	NC_088856.1	NW_026863691.1	1.16e-06	10
43	NC_088856.1	NW_026863406.1	1.16e-06	10
44	NC_088855.1	NW_026865227.1	5.21e-04	10
45	NC_088855.1	NW_026852806.1	4.90e-03	10
46	NC_088860.1	NW_026866838.1	2.04e-05	9
47	NC_088860.1	NW_026851646.1	1.87e-04	9
48	NC_088859.1	NW_026861844.1	2.50e-03	9
49	NC_088859.1	NW_026859085.1	2.50e-03	9
50	NC_088859.1	NW_026856555.1	2.50e-03	9
51	NC_088859.1	NW_026855828.1	2.15e-02	9
52	NC_088859.1	NW_026855648.1	2.50e-03	9
53	NC_088859.1	NW_026851615.1	2.50e-03	9
54	NC_088858.1	NW_026867154.1	1.25e-03	9
55	NC_088858.1	NW_026866080.1	1.09e-02	9
56	NC_088858.1	NW_026865813.1	5.23e-02	9
57	NC_088858.1	NW_026864957.1	1.25e-03	9
58	NC_088858.1	NW_026861602.1	5.23e-02	9
59	NC_088858.1	NW_026856137.1	1.09e-02	9
60	NC_088858.1	NW_026853973.1	1.25e-03	9
61	NC_088858.1	NW_026852645.1	1.09e-02	9
62	NC_088857.1	NW_026864022.1	2.08e-04	9
63	NC_088855.1	NW_026857461.1	3.28e-03	9
64	NC_088855.1	[illegible]	[illegible]	9
72	NC_088860.1	NW_026859162.1	2.26e-04	8
73	NC_088859.1	NW_026866098.1	1.62e-02	8
74	NC_088859.1	NW_026857358.1	1.62e-02	8
75	NC_088859.1	NW_026852039.1	1.62e-02	8
76	NC_088858.1	NW_026866929.1	8.76e-03	8
77	NC_088858.1	NW_026863877.1	8.76e-03	8
78	NC_088857.1	NW_026865475.1	6.43e-02	8

e

Cgig vs Rphi FET, chrom. pieces

Cgig scaffolds

	Cgig scaf	Rphi scaf	p (FET)	Count
0	NC_088859.1 : 0-end	NW_026860903.1 : 0-end	4.30e-13	21
1	NC_088858.1 : 0-end	NW_026865571.1 : 0-end	1.09e-11	20
2	NC_088858.1 : 0-end	NW_026865484.1 : 0-end	6.64e-10	19
3	NC_088858.1 : 0-end	NW_026861563.1 : 0-end	7.31e-11	19
4	NC_088859.1 : 0-end	NW_026862663.1 : 0-end	1.66e-08	18
5	NC_088858.1 : 0-end	NW_026861019.1 : 0-end	2.57e-08	18
6	NC_088860.1 : 0-end	NW_026851633.1 : 0-end	1.49e-11	16
7	NC_088859.1 : 0-end	NW_026853802.1 : 0-end	5.08e-09	16
8	NC_088859.1 : 0-end	NW_026852008.1 : 0-end	4.54e-07	15
9	NC_088858.1 : 0-end	NW_026852258.1 : 0-end	1.06e-06	15
10	NC_088856.1 : 0-end	NW_026858503.1 : 0-end	1.61e-13	15
11	NC_088855.1 : 0-end	NW_026863896.1 : 0-end	2.63e-05	15
12	NC_088855.1 : 0-end	NW_026855197.1 : 0-end	1.06e-04	15
13	NC_088859.1 : 0-end	NW_026865967.1 : 0-end	2.77e-06	14
14	NC_088859.1 : 0-end	NW_026859933.1 : 0-end	9.22e-05	14
15	NC_088859.1 : 0-end	NW_026854960.1 : 0-end	2.16e-07	14
16	NC_088859.1 : 0-end	NW_026852705.1 : 0-end	2.77e-06	14
17	NC_088858.1 : 0-end	NW_026863541.1 : 0-end	7.32e-08	14
18	NC_088859.1 : 0-end	NW_026853170.1 : 0-end	1.40e-06	13
19	NC_088857.1 : 0-end	NW_026859561.1 : 0-end	3.84e-08	13
20	NC_088855.1 : 0-end	NW_026861864.1 : 0-end	2.48e-05	13
21	NC_088855.1 : 0-end	NW_026861084.1 : 0-end	1.59e-04	13
22	NC_088854.1 : 0-end	NW_026852000.1 : 0-end	6.37e-07	13
23	NC_088859.1 : 0-end	NW_026862783.1 : 0-end	1.02e-04	12
24	NC_088858.1 : 0-end	NW_026859790.1 : 0-end	3.62e-06	12
25	NC_088855.1 : 0-end	NW_026855294.1 : 0-end	3.68e-02	12
26	NC_088860.1 : 0-end	NW_026863645.1 : 0-end	1.82e-06	11
27	NC_088859.1 : 0-end	NW_026867092.1 : 0-end	5.93e-05	11
28	NC_088858.1 : 0-end	NW_026853905.1 : 0-end	2.54e-05	11
29	NC_088857.1 : 0-end	NW_026861840.1 : 0-end	7.33e-04	11
30	NC_088856.1 : 0-end	NW_026867313.1 : 0-end	7.66e-09	11
31	NC_088855.1 : 0-end	NW_026857770.1 : 0-end	1.88e-02	11
32	NC_088855.1 : 0-end	NW_026857221.1 : 0-end	8.48e-04	11
33	NC_088859.1 : 0-end	NW_026866451.1 : 0-end	3.64e-03	10
34	NC_088859.1 : 0-end	NW_026864352.1 : 0-end	1.88e-02	10
35	NC_088859.1 : 0-end	NW_026859998.1 : 0-end	3.85e-04	10
36	NC_088859.1 : 0-end	NW_026859757.1 : 0-end	3.85e-04	10
37	NC_088858.1 : 0-end	NW_026866837.1 : 0-end	1.71e-03	10
38	NC_088858.1 : 0-end	NW_026862689.1 : 0-end	1.71e-03	10
39	NC_088858.1 : 0-end	NW_026859620.1 : 0-end	1.71e-03	10
40	NC_088858.1 : 0-end	NW_026859418.1 : 0-end	1.78e-04	10
41	NC_088858.1 : 0-end	NW_026852468.1 : 0-end	1.71e-03	10
42	NC_088856.1 : 0-end	NW_026863691.1 : 0-end	1.16e-06	10
43	NC_088856.1 : 0-end	NW_026863406.1 : 0-end	1.16e-06	10
44	NC_088855.1 : 0-end	NW_026865227.1 : 0-end	5.21e-04	10
45	NC_088855.1 : 0-end	NW_026852806.1 : 0-end	4.90e-03	10
46	NC_088860.1 : 0-end	NW_026866838.1 : 0-end	2.04e-05	9
47	NC_088860.1 : 0-end	NW_026851646.1 : 0-end	1.87e-04	9
48	NC_088859.1 : 0-end	NW_026861844.1 : 0-end	2.50e-03	9
49	NC_088859.1 : 0-end	NW_026859085.1 : 0-end	2.50e-03	9
50	NC_088859.1 : 0-end	NW_026856555.1 : 0-end	2.50e-03	9
51	NC_088859.1 : 0-end	NW_026855828.1 : 0-end	2.15e-02	9
52	NC_088859.1 : 0-end	NW_026855648.1 : 0-end	2.50e-03	9
53	NC_088859.1 : 0-end	NW_026851615.1 : 0-end	2.50e-03	9
54	NC_088858.1 : 0-end	NW_026867154.1 : 0-end	1.25e-03	9
55	NC_088858.1 : 0-end	NW_026866080.1 : 0-end	1.09e-02	9
56	NC_088858.1 : 0-end	NW_026865813.1 : 0-end	5.23e-02	9
57	NC_088858.1 : 0-end	NW_026864957.1 : 0-end	1.25e-03	9
58	NC_088858.1 : 0-end	NW_026861602.1 : 0-end	5.23e-02	9
59	NC_088858.1 : 0-end	NW_026856137.1 : 0-end	1.09e-02	9
60	NC_088858.1 : 0-end	NW_026853973.1 : 0-end	1.25e-03	9
61	NC_088858.1 : 0-end	NW_026852645.1 : 0-end	1.09e-02	9
62	NC_088857.1 : 0-end	NW_026864022.1 : 0-end	2.08e-04	9
63	NC_088855.1 : 0-end	NW_026857461.1 : 0-end	3.28e-03	9
64	NC_088855.1 : 0-end	[illegible]	[illegible]	9
72	NC_088860.1 : 0-end	NW_026859162.1 : 0-end	2.26e-04	8
73	NC_088859.1 : 0-end	NW_026866098.1 : 0-end	1.62e-02	8
74	NC_088859.1 : 0-end	NW_026857358.1 : 0-end	1.62e-02	8
75	NC_088859.1 : 0-end	NW_026852039.1 : 0-end	1.62e-02	8
76	NC_088858.1 : 0-end	NW_026866929.1 : 0-end	8.76e-03	8
77	NC_088858.1 : 0-end	NW_026863877.1 : 0-end	8.76e-03	8
78	NC_088857.1 : 0-end	NW_026865475.1 : 0-end	6.43e-02	8

(a) depicts the chromosomes/scaffolds of Cgig (x-axis) plotted against the chromosomes/scaffolds of Rphi (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 10137 orthologs found between 12 Cgig scaffolds and 4167 Rphi scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. Nature Ecology & Evolution 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chormosome FET p-value , and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.